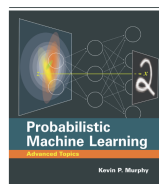
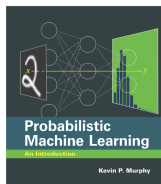
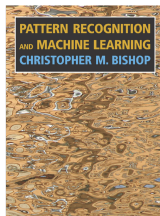
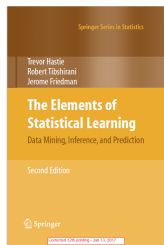


Statistical and Machine Learning

Introduction

Katarzyna Reluga

Literature



- ▶ The Elements of Statistical Learning, 2nd edition (ESLII)
- ▶ Pattern Recognition and Machine Learning (PRML)
- ▶ Probabilistic Machine Learning (PML)

Thanks

- ▶ To Prof. Dr. Xiangnan Xu for sharing the slides from the past edition of the course.
- ▶ To Prof. Dr. Sonja Greven for providing lots of materials.
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- ▶ To Mara Lein for her commitment, dedication and help in the preparation of this year edition of the lecture slides and exercise sessions as well as for the delivery of the exercises sessions.
- ▶ To you for all comments, corrections etc.

Introduction

- ▶ What is machine learning?
- ▶ What is machine learning able to do?
- ▶ What can I learn from this lecture?
- ▶ What cannot I learn from this lecture?

- ▶ **What is machine learning?**
- ▶ What is machine learning able to do?
- ▶ What can I learn from this lecture?
- ▶ What cannot I learn from this lecture?

What is machine learning?

Machine learning

🌐 88 languages ▾

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From Wikipedia, the free encyclopedia

For the journal, see [Machine Learning \(journal\)](#).

"Statistical learning" redirects here. For statistical learning in linguistics, see [Statistical learning in language acquisition](#).

Machine learning (ML) is a [field of study](#) in [artificial intelligence](#) concerned with the development and study of [statistical algorithms](#) that can learn from [data](#) and [generalize](#) to unseen data, and thus perform [tasks](#) without explicit programming language [instructions](#).^[1] Within a subdiscipline of machine learning, advances in the field of [deep learning](#) have allowed [neural networks](#), a class of statistical algorithms, to surpass many previous machine learning approaches in performance.

[Statistics](#) and [mathematical optimisation](#) (mathematical programming) methods compose the foundations of machine learning. [Data mining](#) is a related field of study, focusing on [exploratory data analysis](#) (EDA) through [unsupervised learning](#).^{[3][4]}

From a theoretical viewpoint, [probably approximately correct learning](#) provides a mathematical and statistical framework for describing machine learning. Most traditional machine learning and deep learning algorithms can be described as [empirical risk minimisation](#) under this framework.

Part of a series on

Machine learning and data mining

Paradigms [\[show\]](#)

Problems [\[show\]](#)

Supervised learning [\[show\]](#)
([classification](#) · [regression](#))

Clustering [\[show\]](#)

Dimensionality reduction [\[show\]](#)

Structured prediction [\[show\]](#)

Anomaly detection [\[show\]](#)

Neural networks [\[show\]](#)

Reinforcement learning [\[show\]](#)

Learning with humans [\[show\]](#)

What is machine learning?

A popular definition of **machine learning** or **ML**, due to Tom Mitchell [Mit97], is as follows:

A computer program is said to learn from experience E with respect to some class of tasks T , and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .

Almost all of machine learning can be viewed in probabilistic terms, making probabilistic thinking fundamental. It is, of course, not the only view. But it is through this view that we can connect what we do in machine learning to every other computational science, whether that be in stochastic optimisation, control theory, operations research, econometrics, information theory, statistical physics or bio-statistics. For this reason alone, mastery of probabilistic thinking is essential.

By Shakir Mohamed, researcher at Deep Mind.

What is machine learning?

ML develops **algorithms** (“machines”) that **learn** to perform **tasks** from **data** (“past experience”).



$$\text{SML} = \text{ML} + \text{STAT}$$

- ▶ What is machine learning?
- ▶ **What is machine learning able to do?**
- ▶ What can I learn from this lecture?
- ▶ What cannot I learn from this lecture?

Example 1: Iris data

- Task – prediction of labels



index	sl	sw	pl	pw	label
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
...					
50	7.0	3.2	4.7	1.4	Versicolor
...					
149	5.9	3.0	5.1	1.8	Virginica

input space, $\mathcal{X}$ = numeric features

$$\mathcal{X} = \mathbb{R}^4$$

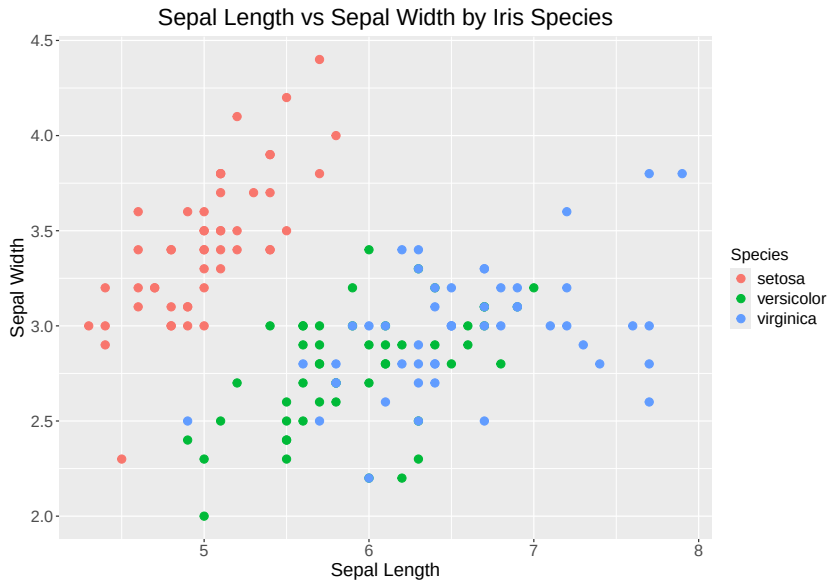
$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

Data source: iris R

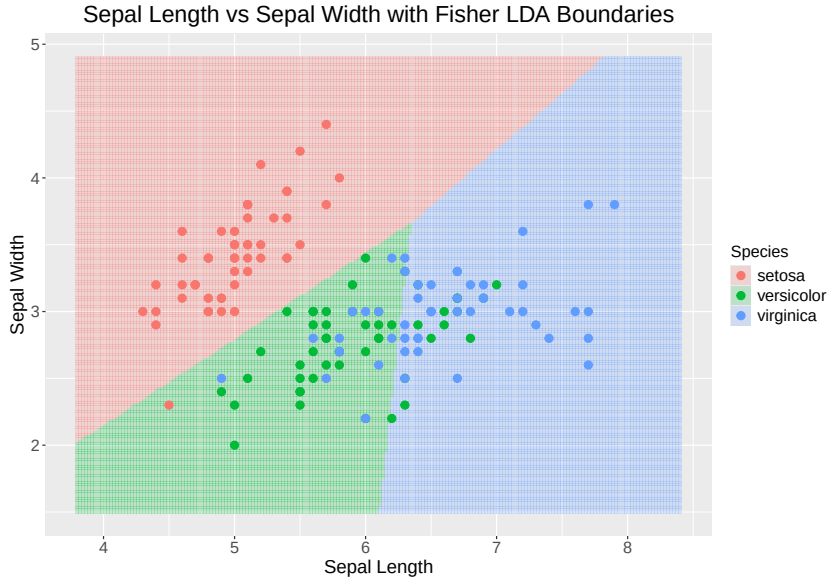
Notation

- ▶ In most cases in this course, we will denote by $\mathcal{X}$ the input space, by $\mathcal{Y}$ the output/response space, and by f a function from $\mathcal{X}$ to $\mathcal{Y}$.
- ▶ Uppercase letters $X, Y, Z, \dots$ denote random variables, and lowercase letters $x, y, z, \dots$ denote realizations or observed values.
- ▶ n denotes the sample size, that is, the number of observed samples, and p denotes the number of variables or features.
- ▶ Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote a data matrix, where each row represents a sample and each column represents a feature or variable.
- ▶ Features = Covariates = Variables = Dimensions of $\mathcal{X}$
- ▶ Samples = Observations = Rows of $\mathbf{X} \in \mathbb{R}^{n \times p}$

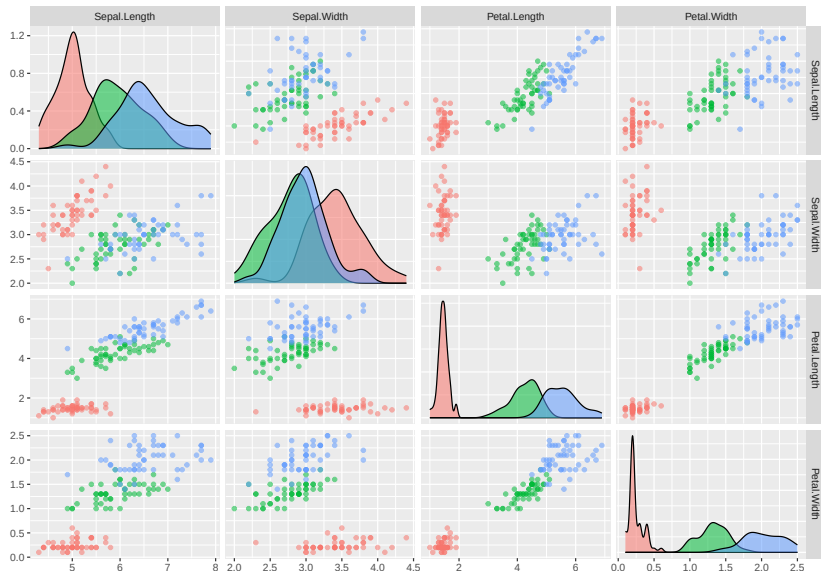
Example 1: Iris data



Example 1: Iris data – classification



Example 1: Iris data – scatter plot



Questions

1. How can we learn classification rules from a given data set?
2. Given new data, how can we classify new samples or observations?
3. Is the classifier optimal with respect to a given criterion?
4. How can we make the best use of all four variables?
5. Should we use a linear or a nonlinear classifier?

Example 2: Image data

- ▶ **Task** – recognition of handwritten digits

label = 5



label = 0



label = 4



label = 2



label = 1



label = 3



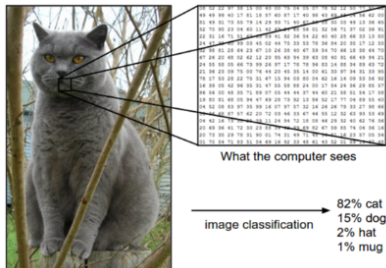
input space, $\mathcal{X} =$
set of handwritten digits images
 $\mathcal{X} = \mathbb{R}^D$, where $D = D_1 \times D_2$

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

- ▶ These digits are given to us as small digital images.
 - ▶ We must build a “machine” that decides which digit is shown.
 - ▶ Challenge: people write digits in many different ways, so this is considered a more complex task.

Data source: MNIST data

General image classification



input space, $\mathcal{X}$ = set of images
 $\mathcal{X} = \mathbb{R}^D$, where $D = C \times D_1 \times D_2$

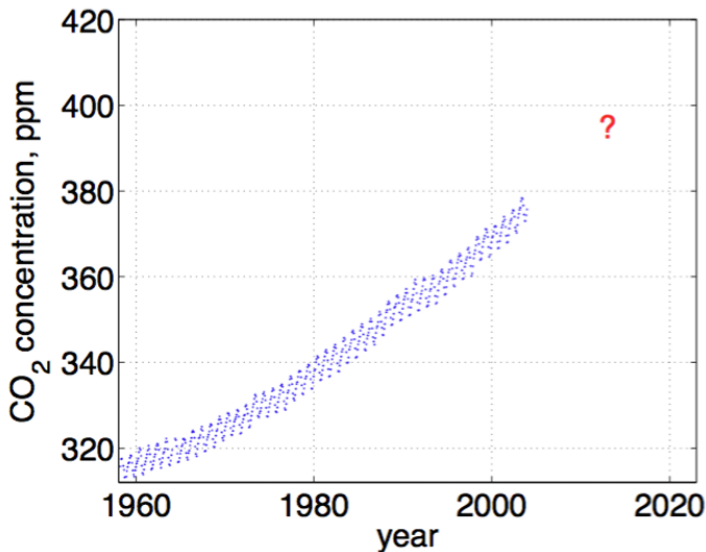
$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

Data source: ImageNet

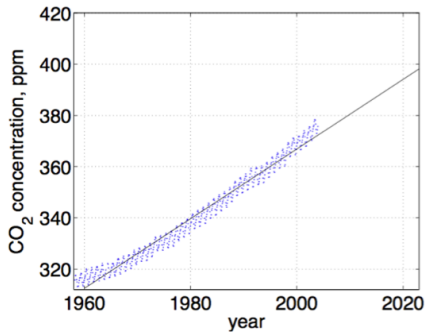
Quick Summary

- ▶ A general framework
 - ▶ Given a set of data $\mathcal{X}$ and their labels $\mathcal{Y}$ (discrete), learn a classification function $f : \mathcal{X} \rightarrow \mathcal{Y}$
- ▶ What kind of methods will you learn?
 - ▶ Logistic regression
 - ▶ K-nearest neighbour
 - ▶ Linear discriminant analysis
 - ▶ Support vector machines
 - ▶ Random forest
 - ▶ Boosting
 - ▶ Neural networks
 - ▶ ...

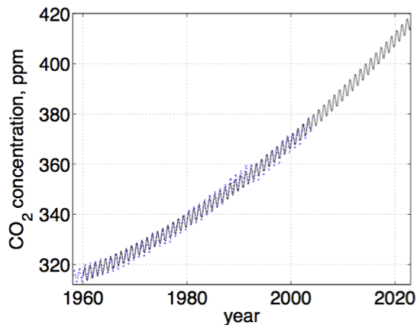
Example 3: CO₂ prediction



Example 3: CO₂ prediction – choice of the model



Linear predictor

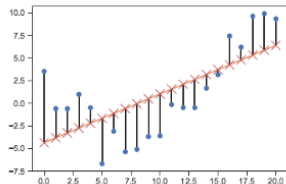


More complex model

Choice of the model or feature engineering?

Linear Regression

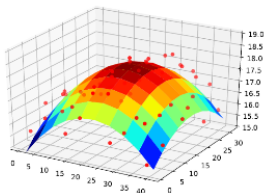
1 Feature



$$f(x; \theta) = b + wx$$

Polynomial Regression

Features' Engineering



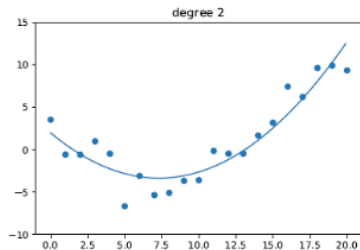
$$f(x; w) = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2$$

Deep Neural Network

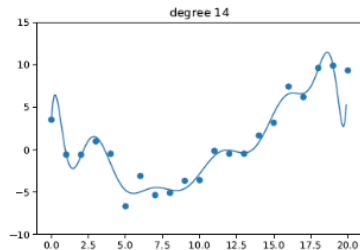
Feature
Extraction
Automatically

$$f(x; w, V) = w^\top \phi(x; V)$$

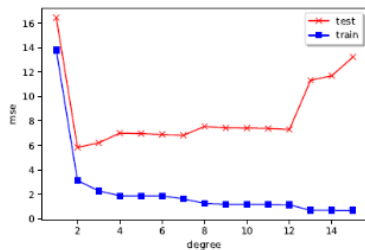
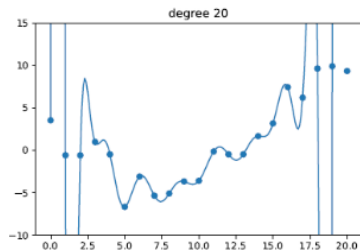
Key challenge: generalization



(a)



(b)



Further challenges and questions

- ▶ How should we proceed when we have many variables, that is, nearly as many as or more than the number of observations?
- ▶ Which variables should we use?
- ▶ What kind of nonlinear model should we use?
- ▶ How can we incorporate prior knowledge into the model? Should we do so?
- ▶ How can we avoid overfitting and improve predictive performance?
- ▶ When can we generalize our results? How can we use the constructed learner, predictor, or model?

Summary of Supervised Learning

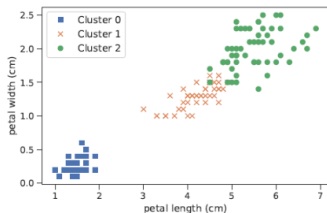
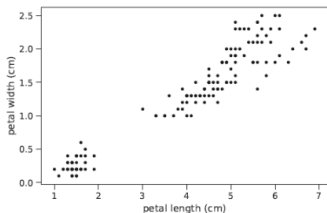
- ▶ A general framework: given a set of data $\mathcal{X}$ and their labels/responses $\mathcal{Y}$ (discrete/continuous/count/...), learn a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$.
- ▶ Methods you will learn:
 - ▶ (Bayesian) Linear regression
 - ▶ Support vector regression
 - ▶ Random forest/Boosting
 - ▶ Neural networks
 - ▶ ...
- ▶ **Can we find a single method that works for all cases?**

- ▶ No free lunch theorem (NFL) by Wolpert and Macready (1996)

Theorem 1: Given a finite set $\mathcal{V}$ and a finite set $\mathcal{S}$ of real numbers, assume that $f : \mathcal{V} \rightarrow \mathcal{S}$ is chosen at random according to uniform distribution on the set $\mathcal{S}^{\mathcal{V}}$ of all possible functions from $\mathcal{V}$ to $\mathcal{S}$. For the problem of optimizing f over the set $\mathcal{V}$, no algorithm performs better than blind search.

- ▶ Unfortunately, there is **no single best model** that works optimally for all kinds of problems.
- ▶ The reason is that a set of assumptions (also called inductive bias) that works well in one domain/for a specific problem may work poorly in another.
- ▶ When selecting the best model, one should clearly define the optimality criterion using domain knowledge and the specifics of the learning task.

Example 4: How can we partition the input space into regions containing “similar” points?

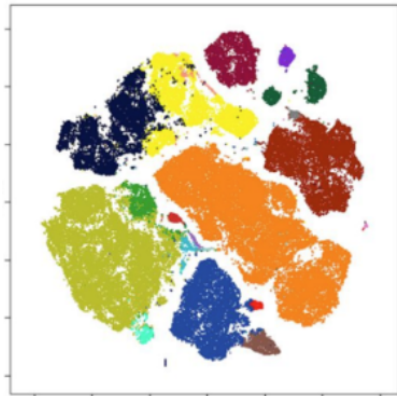


Key feature: we know nothing about the label in the data, i.e. we don't have $\mathcal{Y}$.

Example 5: How can we present/visualize high dimensional data set?

Table

	Feature 1	Feature 2	Feature 3	
Sample 1	2	0	1	...
Sample 2	0	1	0	...
Sample 3	1	0	0	...
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.



Key feature: we know nothing about the label in the data, i.e. we don't have $\mathcal{Y}$.

Summary of Unsupervised learning

- ▶ A general framework: given a dataset $\mathcal{X}$, find a function f that maps $\mathcal{X}$ to a lower-dimensional space that provides a good representation of the original data while preserving as much of the original information in $\mathcal{X}$ as possible.
- ▶ Methods you will learn?
 - ▶ Principal component analysis (PCA)
 - ▶ k-means/hierarchical clustering
 - ▶ Model based clustering
 - ▶ ...

- ▶ What is machine learning?
- ▶ What is machine learning able to do?
- ▶ **What can I learn from this lecture?**
- ▶ What cannot I learn from this lecture?

What can I learn from this lecture?

Supervised:

1. Linear models
2. Statistical regularization (Lasso and variants)
3. Tree-based model (Decision tree, Random forest)
4. Boosting (Adaboost, Gradient boosting)
5. Kernel method (SVM)
6. Neural Networks I (CNN/RNN/LSTM)

Unsupervised:

7. Clustering (hierarchical, model-based)
8. Dimension reduction (PCA, manifold learning)
9. Neural Networks II (AE/VAE)

Why should I learn about the methods covered in this course?

🏠 > Data > AI and Machine Learning > 10 Machine Learning Algorithms to Know in 2026

10 Machine Learning Algorithms to Know in 2026

Written by Coursera Staff • Updated on Dec 4, 2025

🔗 Share

Machine learning algorithms power many services in the world today. Here are 10 to know as you look to start your career.



1. **Linear regression**
2. **Logistic regression**
3. **Naive Bayes**
4. **Decision tree**
5. **Random forest**
6. **K-nearest neighbour (KNN)**
7. **K-means**
8. **Support vector machine**
9. **Apriori**
10. **Gradient boosting**

Is this course useful for me?

- ▶ I would like to find a job after my graduation
 - ▶ Example: Data Scientists in BCG X:

We are looking for dedicated individuals with a passion for data science, statistics, operations research and redefining organizations into AI led innovative companies. Successful candidates possess the following:

- Comfortable in a client-facing role with the ambition to lead teams

- Likes to distill complex results or processes into simple, clear visualizations

- Explain sophisticated data science concepts in an understandable manner

- Love building things and are comfortable working with modern development tools and writing code collaboratively (bonus points if you have a software development or DevOps experience)

- Significant experience applying advanced analytics to a variety of business situations and a proven ability to synthesize complex data

- Deep understanding of modern machine learning techniques and their mathematical underpinnings, and can translate this into business implications for our clients

- Have strong project management skills

Is this course useful for me?

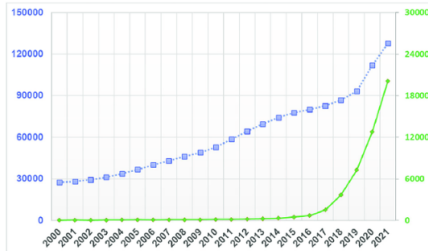
- I would like to do research (Ph.D. or similar) in the future

Gartner Hype Cycle for Emerging Technologies, 2016



gartner.com/SmarterWithGartner

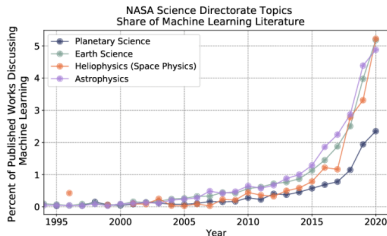
Gartner.



Temporal trends in machine learning-centered articles published in PubMed between 2000 and 2021. This figure shows the results of a simple search on PubMed for clinical diagnostic studies focused on 'diagnosis' between 2000 and 2021 (line with square markers) and the subportion of those studies that reference 'machine learning', 'artificial intelligence', or 'deep learning' in the title or abstract (line with diamond markers).

Is this course useful for me?

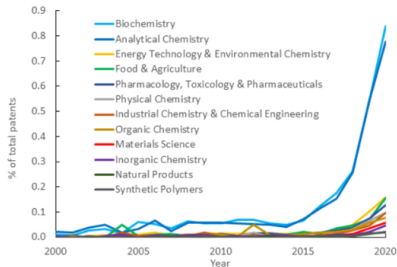
Planetary Science



Source:

<https://arxiv.org/abs/2007.15129>

Chemical Science



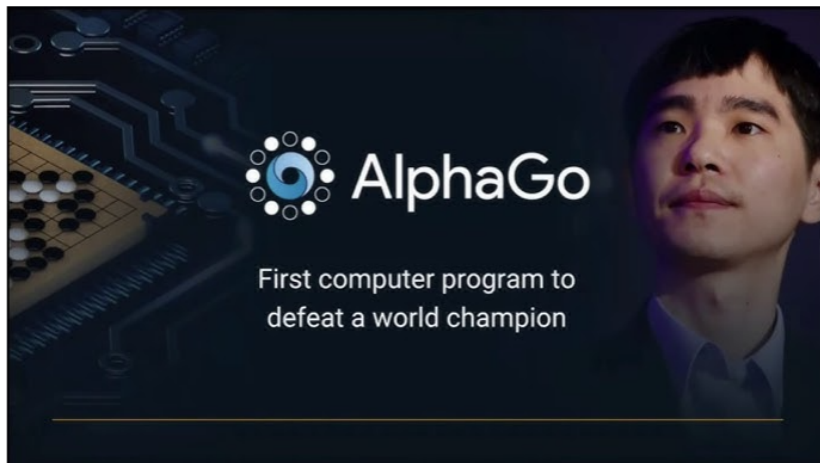
Source:

<https://pubs.acs.org/doi/10.1021/acs.jcim.1c00619>

- ▶ What is machine learning?
- ▶ What is machine learning able to do?
- ▶ What can I learn from this lecture?
- ▶ **What cannot I learn from this lecture?**

What cannot you expect from this course?

- ▶ Reinforcement learning – learn how to interact with the environment by applying a policy which specifies which action to take in response to each possible input (derived from the environment state).



Machine Learning is still on the way

- ▶ How can we achieve better generalization?
- ▶ How can we interpret machine learning results?
- ▶ How can we incorporate causality into machine learning?
- ▶ To what extent can machine learning truly mimic human reasoning?
- ▶ How can we integrate machine learning with scientific discovery?
- ▶ ...

The world is waiting for you! :)